

3 - Evaluation by measured spectra

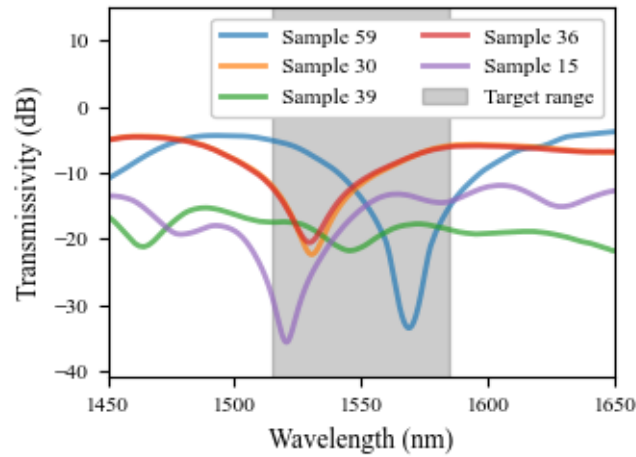
October 8, 2024

1 Load the measured data

Load the measured dataset (LPFG spectra) and the same spectra preprocessed by simulated FBGs

1.1 Visualize some transmission spectra

Below some measured spectra are illustrated along side the proposed interrogation system's measurement range.



2 Load the model

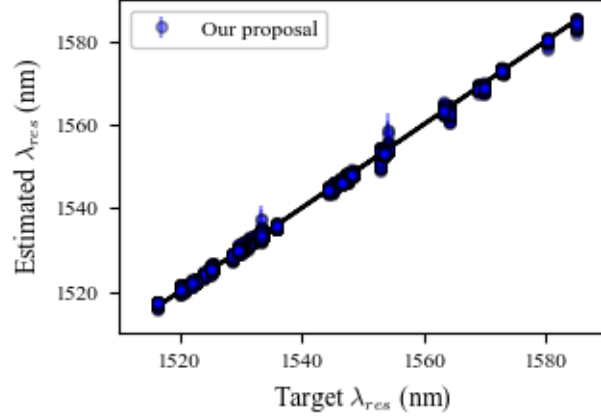
Load the LPFG decoder model and the fine tuned weights

3 Model evaluation

3.1 Proposed model predictions

Run the model using the measured spectra for several FBG positions.

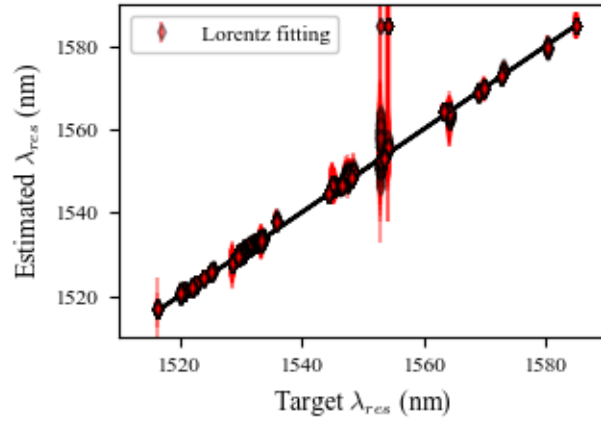
Below one can see this models output compared to the target LPFG position.



3.2 Base model predictions

Run the baseline model for the same samples.

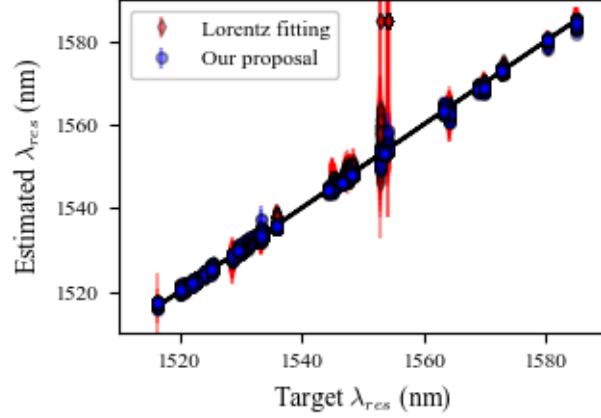
Below one can see this models output compared to the target LPFG position.



3.3 Comparison

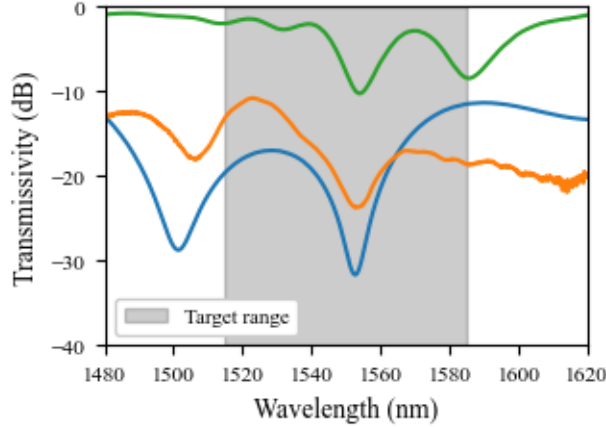
We found that the proposed model performs better than the baeline model (Loretzian fitting).

The performace improvement is more prominent for some spectra, we searched for them and tried to explain this extensive difference.



3.3.1 Lorentzian prediction outliers

Below one can find the spectra in which the Lorentz fitting provided a poor estimate.



Note that for spectra with distortions and some ripples, the Lorentzian fit model had lower performance, specially when the sensor presents a second resonant wavelength outside the target interrogator range.

We believe that the incorporation of the attention mechanism could have improved this problem. This assumption was made because the attention mechanism might suppress the features related to the second resonant wavelength and other distortions, not related to the desired resonant wavelength.

3.4 Performance metrics

Below one can find the performance metrics for both approaches, the proposed self-attention neural net and the baseline Lorentzian fit.

We also restricted the dataset regarding the spectra which the Lorentzian fit was too poor.

	full dataset			\
	rmse	mape	r2	
Proposed model	0.738+/-0.013	0.03343+/-0.00015	0.99783+/-0.00008	
Loretzian fitted model	2.13+/-0.35	0.0500+/-0.0022	0.982+/-0.006	

	no outliers			
	rmse	mape	r2	
Proposed model	0.686+/-0.009	0.03207+/-0.00015	0.99814+/-0.00005	
Loretzian fitted model	0.83+/-0.04	0.0392+/-0.0015	0.99731+/-0.00027	

Subset: full dataset, Metric: rmse

T-statistic: -337.62

P-value: 0.000

There is a statistically significant difference between the rmse means.

Subset: full dataset, Metric: mape

T-statistic: -642.14

P-value: 0.000

There is a statistically significant difference between the mape means.

Subset: full dataset, Metric: r2

T-statistic: 227.44

P-value: 0.000

There is a statistically significant difference between the r2 means.

Subset: no outliers, Metric: rmse

T-statistic: -278.41

P-value: 0.000

There is a statistically significant difference between the rmse means.

Subset: no outliers, Metric: mape

T-statistic: -400.43

P-value: 0.000

There is a statistically significant difference between the mape means.

Subset: no outliers, Metric: r2

T-statistic: 256.30

P-value: 0.000

There is a statistically significant difference between the r2 means.

4 Refractive index sensor

An LPFG sensor was developed and characterized regarding RI changes using different concentrations of glycerol/water mixture.

4.1 OSA calibration

First, the sensor was characterized using an OSA Anritsu MS9740B. The transmission spectra was recorded in air, water and for the several RI solutions.

4.1.1 The sensor transmission

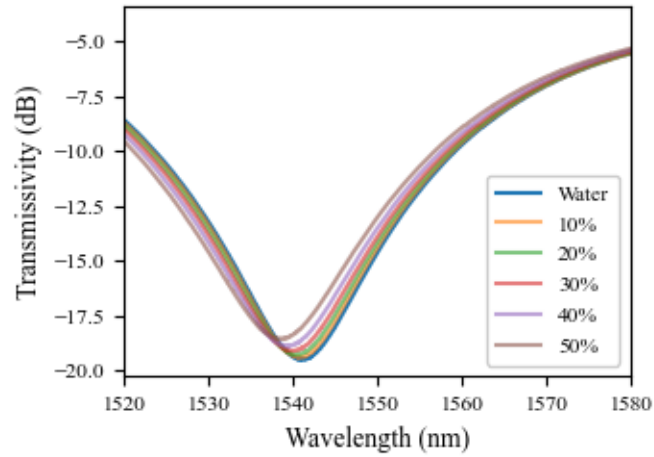
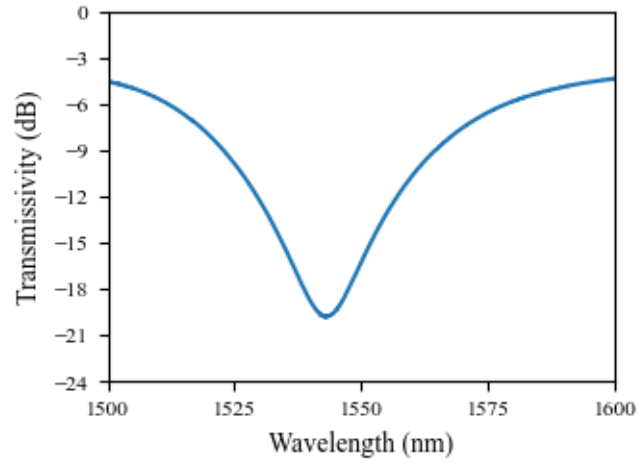
Below some spectral characteristics of the sensor used and its transmission spectra in air.

The spectral response to different glycerol/water mixtures are also provided.

Insertion loss: -4.33 dB

Attenuation: 15.45 dB

FWHM: 52.92 nm

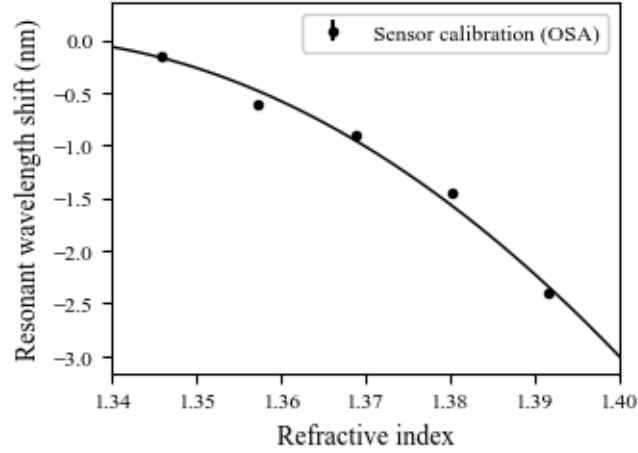


4.1.2 Calibration curve

Calibrate the resonant wavelength regarding the shift from the water sample. This is done to improve temperature crosssensitivity.

$$dwl = -5.80e+02 n^2 + 1.54e+03 n + -1.02e+03$$

Uncertainty in the coefficients: [69.31138525 188.42544222 128.02930423]

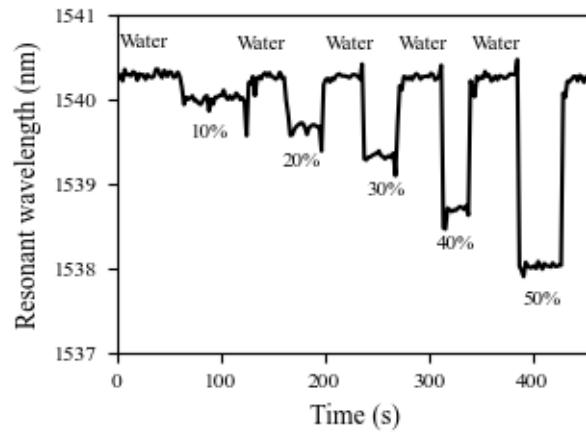


4.2 Measurements using the proposed model

A system was developed to acquire the reflection spectra read by BraggMeter. It extracted the bragg positions and peak intensities, then implemented the proposed interrogation model to estimate the LPFG position. All data was recorded in a database using TinyDB.

Below the measurements time series is shown, with sample alternance. The difference between the water sample used before the RI sample was considered for RI estimation.

10%: 1540.01 $\pm$ 0.02
 20%: 1539.67 $\pm$ 0.03
 30%: 1539.34 $\pm$ 0.01
 40%: 1538.71 $\pm$ 0.01
 50%: 1538.04 $\pm$ 0.01



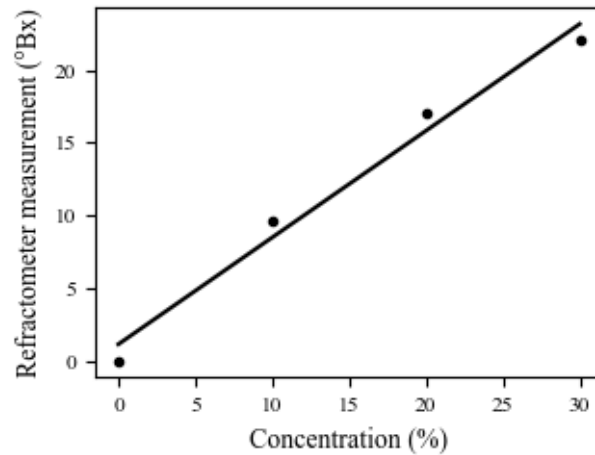
Using the calibration curve, we could estimate the sample RI using the proposed interrogation

method:

Estimated RI: [1.3506403713922608+/-0.0010666223754118578,
1.3602877567416232+/-0.0011250791398116462,
1.368355259183752+/-0.000475062095087122,
1.3798669025522599+/-0.0003700639593616548,
1.3898205176353404+/-0.00043938103969933033]

Below we estimate each sample actual RI. We used an Abbe refractometer with Brix scale, so the scale should be transformed to RIU. Moreover, the 40% and 50% concentrations were outside the refractometer range, so we extrapolated the values using linear fit.

The concentration to brix curve is:



Hence, the sample RI were:

Sample RI

{0: '1.33211+/-0.00016', 10: '1.34751+/-0.00018', 20: '1.35938+/-0.00022', 30:
'1.36740+/-0.00026', 40: '1.38103+/-0.00032', 50: '1.3928+/-0.0004'}

Given the estimated RI and the sample RI, we could calculate the RI errors:

RI perc err: -1.40e-02%

RI bias: -1.68e-04

RI standard error: 1.04e-03

Comparing the sample RI with the estimated RI we could note that most error occurs for lower RI, where the sensitivity is smaller.

